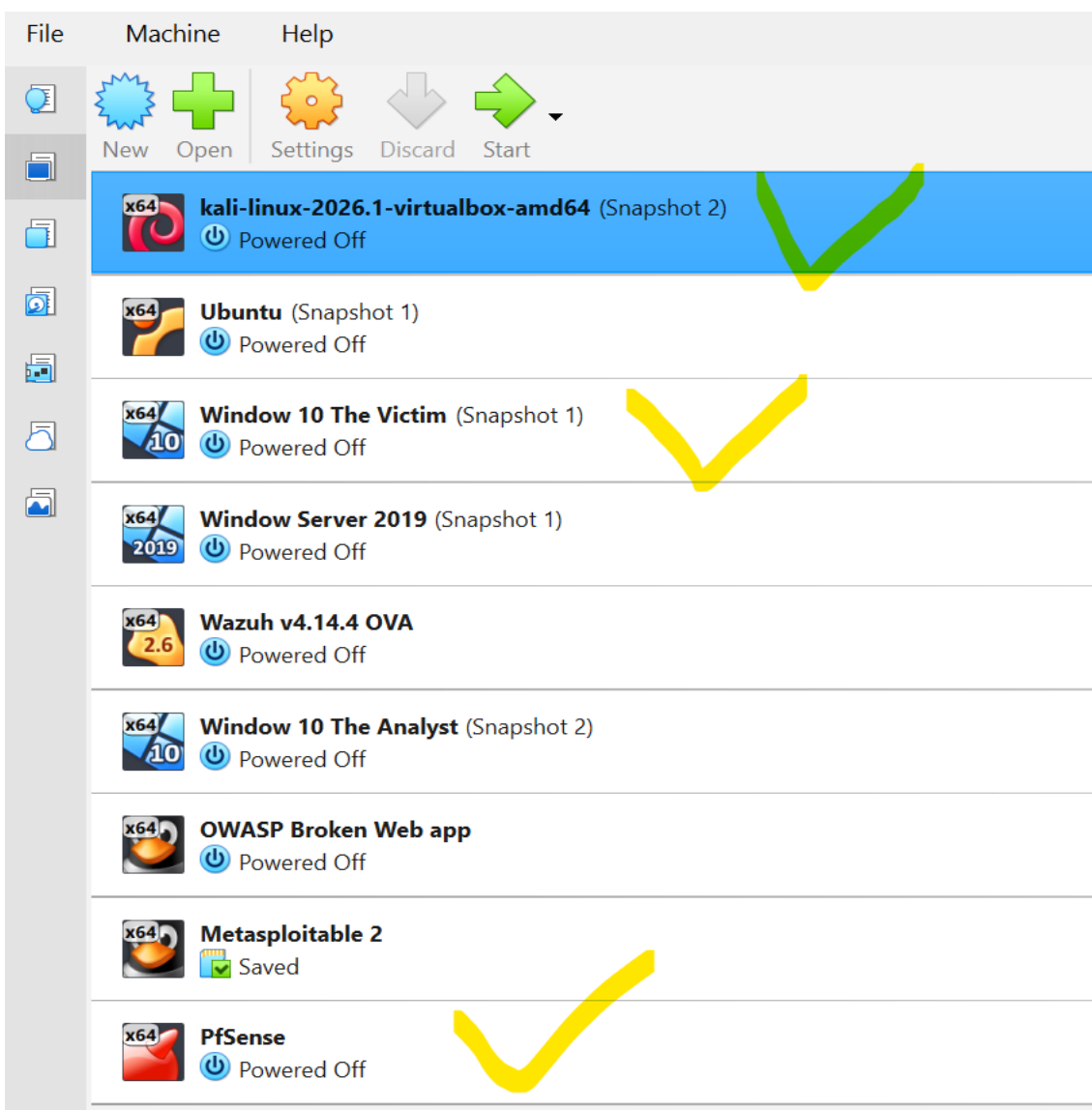


External Network Scanning & Firewall Validation Report

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1. Excutive Summary

This report documents the security assessment and firewall policy validation of an internal Windows 10 target. The objective was to simulate an external attacker (Kali Linux) attempting to perform reconnaissance on a target protected by a pfSense firewall. The lab demonstrates the transition from a permissive security posture to a hardened environment, validated through Nmap scanning and firewall log analysis.



2. Network Topology & Lab Setup

The environment was architected to simulate a real-world DMZ/Internal network split. The pfSense firewall acts as the gateway between the external WAN and internal LAN.

Architecture: [Kali Linux (Attacker)] <-> [pfSense Firewall] <-> [Windows 10 (Target)]			
Device	Interface Mode	Network Segment	IP Address
Kali Linux	Bridged	External WAN	10.128.120.200
pfSense (WAN)	Bridged	External WAN	10.128.120.64
pfSense (LAN)	Internal	LAN Subnet	192.168.1.1
Windows 10	Internal	LAN Subnet	192.168.1.100

*** Welcome to pfSense 2.6.0-RELEASE (amd64) on pfSense ***
WAN (wan) -> em0 -> v4/DHCP4: 10.128.120.64/24
LAN (lan) -> em1 -> v4: 192.168.1.1/24

3. Phase 1: Permissive State (Allow All)

Initially, the pfSense firewall was configured with a "Pass" rule for all incoming traffic. This allows the external attacker to perform full service discovery on the internal host.

Floating WAN LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	4	IPv4*	*	*	*	*	none		/205 KiB	

↑ Add

↓ Add

🗑 Delete

💾 Save

➕ Separator

3.1 External Reconnaissance (Before blocking Firewall Rule)

Step 1: Host Discovery

Command: nmap -sn 192.168.1.100

Step 2: TCP SYN Scan

Command: nmap -sS -p- 192.168.1.100

Step 3: Firewall Detection

Command: nmap -sA 192.168.1.100

```
(kali@kali)-[~]
$ nmap -sn 192.168.1.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 00:44 -0400
Nmap scan report for 192.168.1.100
Host is up (0.00093s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.55 seconds

(kali@kali)-[~]
$ nmap -sS -p- 192.168.1.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 00:45 -0400
Nmap scan report for 192.168.1.100
Host is up (0.0012s latency).
Not shown: 65520 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5040/tcp   open  unknown
5357/tcp   open  wsddapi
7680/tcp   open  pando-pub
49664/tcp  open  unknown
49665/tcp  open  unknown
49666/tcp  open  unknown
49667/tcp  open  unknown
49668/tcp  open  unknown
49669/tcp  open  unknown
49671/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 33.42 seconds

(kali@kali)-[~]
$ nmap -sA 192.168.1.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 00:46 -0400
Nmap scan report for 192.168.1.100
Host is up (0.00093s latency).
All 1000 scanned ports on 192.168.1.100 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 21.83 seconds
```

Findings

Host is up: Connectivity is confirmed.

Open Ports: You can see everything from HTTP (80) and SSH (22) to Windows RPC services.

nmap -sA: Showing 1000 filtered ports with an ACK scan is a great technical touch—it proves the firewall is stateful and is currently dropping packets that aren't part of an established session (standard behavior).

4. Phase 2: Security Policy Enforcement (Deny/Block)

To secure the target, a specific block rule was implemented on the pfSense WAN interface. The rule targets the Kali IP as the source and the Windows 10 IP as the destination.

4.1 Firewall Rule Configuration

- **Action:** Block (Silent Drop)
- **Interface:** WAN
- **Source IP:** 10.128.120.200
- **Destination:** 192.168.1.100

Floating **WAN** LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗ 0 / 0 B	IPv4 *	10.128.120.200	*	192.168.1.100	*	*	none		ip kali block	    
☐	✓ 0 / 5.60 MiB	IPv4 *	*	*	*	*	*	none			    

 Add  Add  Delete  Save  Separator

4.2 Validation Scan

The scan was repeated using the `-Pn` flag (Treat host as online) since ICMP pings were also blocked by the new policy. The results transitioned from **Open** to **Filtered**.

Command: `nmap -Pn -sS 192.168.1.100`

```
(kali㉿kali)-[~]
$ nmap -Pn -sS 192.168.1.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 00:52 -0400
Nmap scan report for 192.168.1.100
Host is up.
All 1000 scanned ports on 192.168.1.100 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 201.91 seconds

(kali㉿kali)-[~]
$ nmap -sS 192.168.1.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 00:57 -0400
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.09 seconds
```

```
(kali㉿kali)-[~]
$ nmap -p 80,22,443,135,139 192.168.1.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 00:59 -0400
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.09 seconds

(kali㉿kali)-[~]
$ nmap -Pn -p 80,22,443,135,139 192.168.1.100
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-07 00:59 -0400
Nmap scan report for 192.168.1.100
Host is up.

PORT      STATE    SERVICE
22/tcp    filtered ssh
80/tcp    filtered http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
443/tcp   filtered https

Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds
```

Findings

All ports appeared as filtered or showed no response.

This indicates that the firewall is successfully blocking incoming traffic.

5. phase 3 Evidence Validation (System Log) pfsense firewall

To verify that the firewall was the entity dropping the traffic, the pfSense System Logs were inspected. The logs confirm that the packets reached the WAN interface but were discarded by the "Default Deny" or specific Block rule.

Last 500 Firewall Log Entries. (Maximum 500)					
Action	Time	Interface	Rule	Source	Destination
✗	May 7 04:46:22	WAN	Default deny rule IPv4 (1000000103)	  10.128.120.200:51840	  192.168.1.100:1500
✗	May 7 04:46:22	WAN	Default deny rule IPv4 (1000000103)	  10.128.120.200:51840	  192.168.1.100:1186
✗	May 7 04:46:22	WAN	Default deny rule IPv4 (1000000103)	  10.128.120.200:51840	  192.168.1.100:1271
✗	May 7 04:46:22	WAN	Default deny rule IPv4 (1000000103)	  10.128.120.200:51840	  192.168.1.100:5802
✗	May 7 04:46:22	WAN	Default deny rule IPv4 (1000000103)	  10.128.120.200:51840	  192.168.1.100:4445

6. Conclusion

The lab successfully demonstrated the effectiveness of a stateful packet inspection firewall. By moving from a permissive to a restrictive policy, we reduced the attack surface of the internal Windows 10 host to zero. The use of `nmap` provided external validation, while the pfSense logs provided internal confirmation of policy enforcement.